Endnotes

1 The terms *centrifugal* and *centripetal* force are sometimes used when describing spinning motion. But they are merely labels. Our intent is to understand why spinning motion gives rise to force.

2 There is debate concerning Mach's precise views on the material that follows. Some of his writings are a bit ambiguous and some of the ideas attributed to him arose from subsequent interpretations of his work. Since he seems to have been aware of these interpretations and never offered corrections, some have suggested that he agreed with their conclusions. But historical accuracy might be better served if every time I write "Mach argued" or "Mach's ideas," you read it to mean "the prevailing interpretation of an approach initiated by Mach."

3 While I like human examples because they make an immediate connection between the physics we're discussing and innate sensations, a drawback is our ability to move, volitionally, one part of our body relative to another—in effect, to use one part of our body as the benchmark for another part's motion (like someone who spins one of his arms relative to his head). I emphasize *uniform* spinning motion—spinning motion in which every part of the body spins together—to avoid such irrelevant complications. So, when I talk about your body's spinning, imagine that, like Newton's two rocks tied by a rope or a skater in the final moments of an Olympic routine, every part of your body spins at the same rate as every other.

4 Like the pages in any flip book, the pages in Figure 3.3 only show representative moments of time. This may suggest to you the interesting question of whether time is discrete or infinitely divisible. We'll come back to that question later, but for now imagine that time is infinitely divisible, so our flip book really should have an infinite number of pages interpolating between those shown.